

CloudChainX Guard: ML-Powered Detection And Blocking Of Cloud Supply Chain API Attacks

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Introduction

Modern API-driven supply chains enable scalability and real-time integration, but they also introduce critical security risks where a single compromised API can disrupt entire systems. This threat extends to sensitive domains such as military logistics, where secure coordination of equipment, ammunition, and supplies across distributed units depends heavily on trusted API communication. In such environments, attacks like IAM role abuse, credential theft, and log tampering can severely impact mission readiness and operational continuity. **CloudChainX Guard** delivers a proactive defense using a Random Forest model achieving 96% accuracy and F1-score to detect and block malicious API requests in real time at the API Gateway before they impact operations. Explainable AI ensures transparent and trustworthy decision-making. Aligned with the cybersecurity priorities of the Ministry of Defense Saudi Arabia, this solution strengthens the protection of critical infrastructure in the Kingdom through a scalable and resilient approach.



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Objectives

- Develop a Machine Learning (ML) Model:** Detect and block API-level attacks in cloud supply chains in real time.
- Enhance Explainability with XAI:** Analyze the rationale behind detected attacks to improve reliability and interpretability.
- Evaluate System Performance:** Assess CloudChainX Guard within a simulated cloud supply chain environment.

Methodology

CloudChainX Guard utilizes machine learning to detect and block API-based cloud supply chain attacks in a cloud supply chain environment. Figure 1 illustrates the proposed workflow.

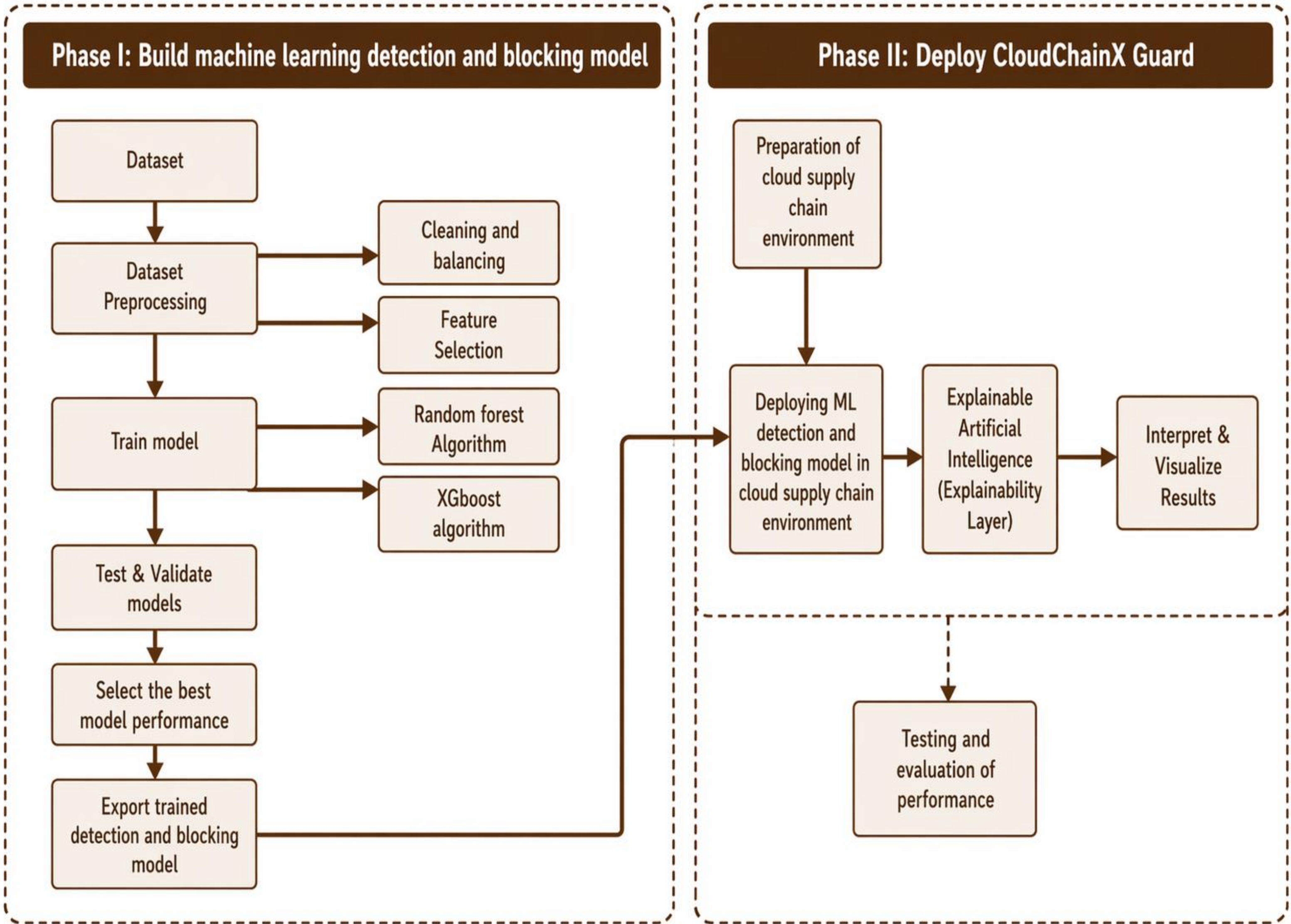


Figure 1: CloudChainX Guard Workflow

Implemntation Overview:

First Phase :

- The “Cloud Attack Dataset Using OWASP” [1], consisting of 10,024 records, was acquired and preprocessed.
- Random Forest and XGBoost models were trained and evaluated using Accuracy, Recall, and F1-score.
- Random Forest was selected as the final model due to its stability and performance.

Second Phase:

- The trained model was deployed in a simulated cloud supply chain environment using Docker and LocalStack.
- API requests were analyzed in real time through the API Gateway.
- Malicious requests were detected and automatically blocked, while normal traffic was allowed.
- Explainable AI (XAI) was integrated to provide interpretable explanations for each detection.
- The system was evaluated against key cloud attack types.

Results and Discussion

Model Performance Results:

Random Forest demonstrated superior classification performance across all evaluation metrics, achieving 96.15% accuracy and 96.12% F1-score, outperforming XGBoost. This reflects its effectiveness in handling structured cloud log data and minimizing misclassification. Figure 2 highlights this performance comparison.

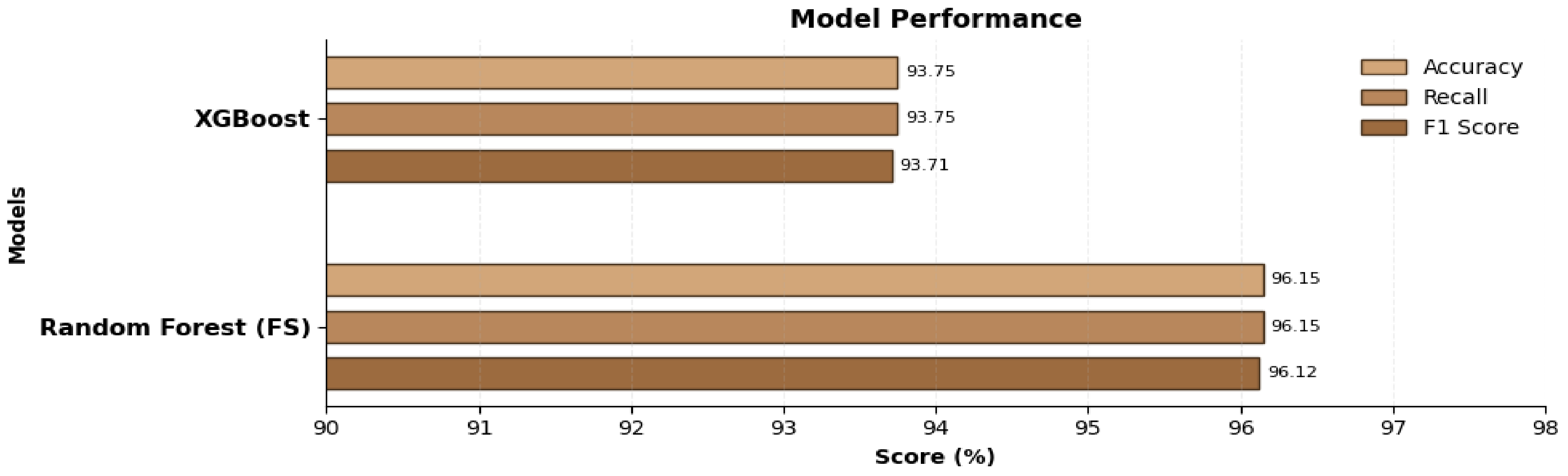


Figure 2: Model Performance Comparison

Detection and Blocking Mechanisms:

Table 1 presents how different cloud supply chain attacks are detected and blocked.

	IAM Role Abuse	Credential Theft	S3 Data Exfiltration	CloudTrail Alteration
Attack Description	Unauthorized use of privileged roles to access resources	Stolen keys/tokens used to impersonate legitimate users	Sensitive data is extracted from cloud storage	Logs are modified or disabled to hide attacks
How Model Detects	Identifies abnormal API behavior and privilege escalation patterns	Identifies unusual login/API patterns and anomalies	Detects unusual data transfer or access patterns	Detects suspicious logging changes or tampering behavior
How Model Blocks	Blocks suspicious role usage before critical actions are executed	Blocks unauthorized access attempts in real time	Prevents data leakage by blocking malicious API requests	Blocks unauthorized attempts to disable or alter logs

Table 1: CloudChainX Guard: Attack Detection & Response Mechanism

Detection, Blocking, and Explainability Results

Figure 3 illustrates real-time monitoring, detected attacks, and XAI explanations.

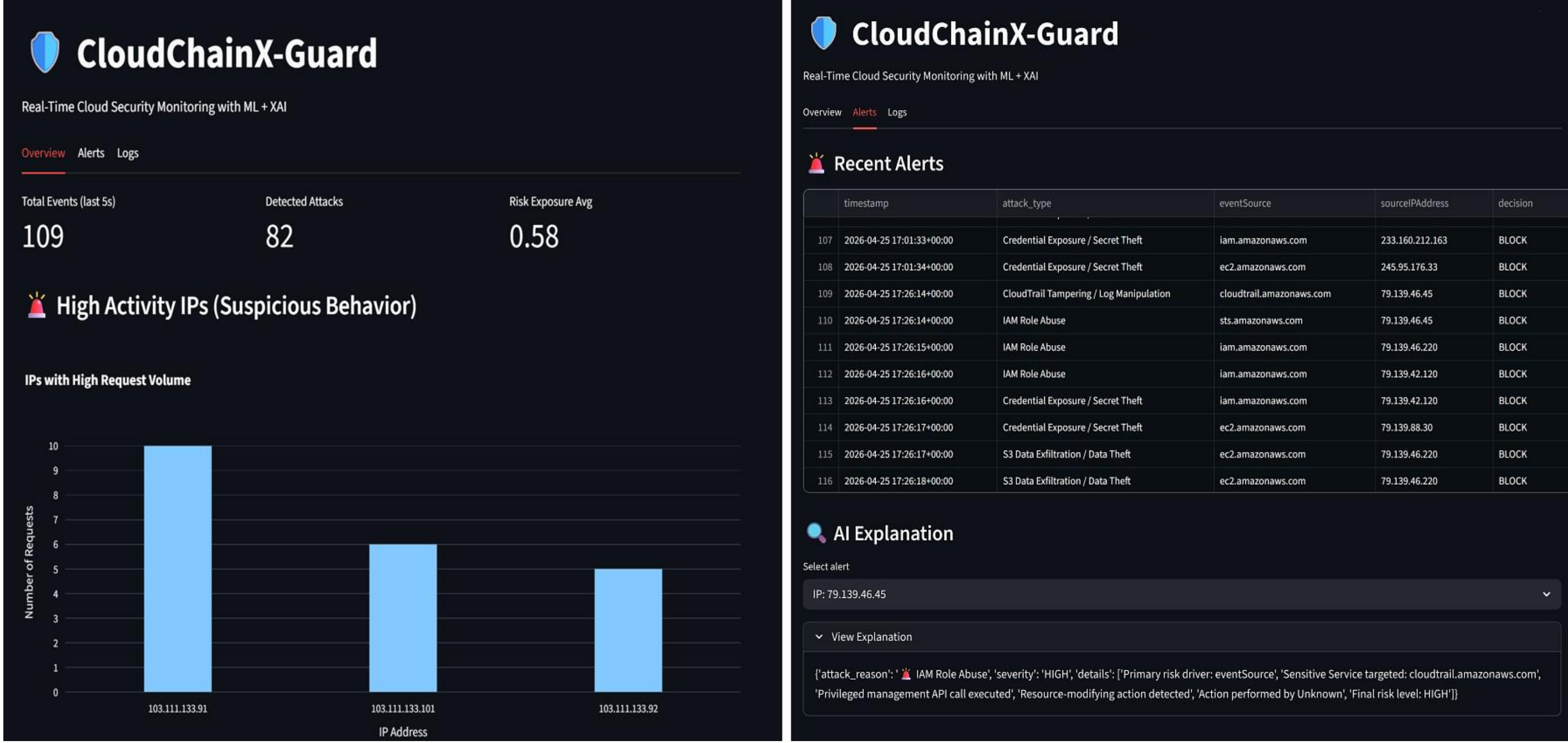


Figure 3: CloudChainX Guard Dashboard

Conclusion

This study demonstrates the effectiveness of CloudChainX Guard in detecting and blocking API-based cloud supply chain attacks in real time. Future work will focus on expanding detection coverage and testing across larger, real-world cloud environments.

Use Case — Military Logistics: Modern military logistics depend on cloud APIs to coordinate the distribution of equipment, ammunition, and supplies across remote units. Such systems are high-value targets for adversaries seeking to disrupt operations through IAM Role Abuse, credential theft, or log tampering. CloudChainX Guard can be deployed to monitor and protect these pipelines in real time, ensuring the integrity and availability of mission-critical supply chain operations.

References

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- [2] J. Kava and Chinmayie, “Cloud Computing a Key to Supply Chain Management: Embracing the Cloud Advantage,” *IJFMR* 23057649, vol. 5, no. 5, 2023, Available: <https://www.ijfmr.com/papers/2023/5/7649.pdf>
- [3] ASMAU OYINLADE OLANIYI, M. G. Adewunmi, and A. Ayeni, “Machine Learning Model for Detecting Attack in Service Supply Chain,” *Journal of Information Security and Cybercrimes Research*, vol. 8, no. 1, pp. 01–16, Jun. 2025, doi: <https://doi.org/10.26735/gnvr4.188>.

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